

National University of Technology

Department of Artificial Intelligence



Usability Detective Case

LAB REPORT 1

Student Name: Aleena Tahir

Student ID: F23607001

Batch: 2023

Instructor: Lec. Murtaza Mehdi

Course: HCI and Computer Graphics Lab (CS341)

Semester: Spring 2026

February 22, 2026

1 Objective

The objective of this lab experiment is to investigate real-world computer interfaces and critically evaluate them through the lens of Human-Computer Interaction (HCI) principles. Acting as usability detectives, we identify design strengths and flaws in two chosen interfaces. The poorly-designed interface is then redesigned using Figma to demonstrate how applying HCI principles leads to improved usability.

- ✓ Evaluate two real-world computer interfaces using HCI principles.
- ✓ Identify strengths in a well-designed interface (Google Calendar).
- ✓ Identify usability failures in a poorly-designed interface (Yale School of Art – `art.yale.edu`).
- ✓ Redesign the flawed interface using Figma, applying HCI improvements.
- ✓ Justify all design decisions with specific HCI theory.

2 Theoretical Background

Human-Computer Interaction (HCI) is a multidisciplinary field focused on the design, evaluation, and implementation of interactive computing systems for human use. Effective interfaces reduce cognitive load, prevent errors, and create satisfying user experiences. Poor interfaces cause frustration, increase error rates, and waste user time. The following key HCI principles serve as the foundation of this analysis:

Table 1: Key HCI Principles

nutDark	
Visibility	All available options and the current system status should be clearly visible to the user.
Feedback	The system should provide immediate, clear responses to user actions (visual, auditory, or haptic).
Affordances	The design of interactive elements should suggest their intended use (e.g., a button looks clickable).
Mapping	Controls should correspond logically to their effects, matching the user's mental model.
Consistency	Similar elements and actions should behave uniformly throughout the interface.
Error Prevention	The interface should minimize errors and provide easy ways to undo or recover from mistakes.

3 Case 1: Well-Designed Interface – Google Calendar

3.1 Interface Overview

Google Calendar (<https://calendar.google.com>) is a widely-used scheduling and time management application developed by Google. It allows users to create, manage, and share events across devices. Its clean layout, color-coded calendars, drag-and-drop interaction, and intuitive weekly/-monthly views make it an exemplary case of user-centered design.

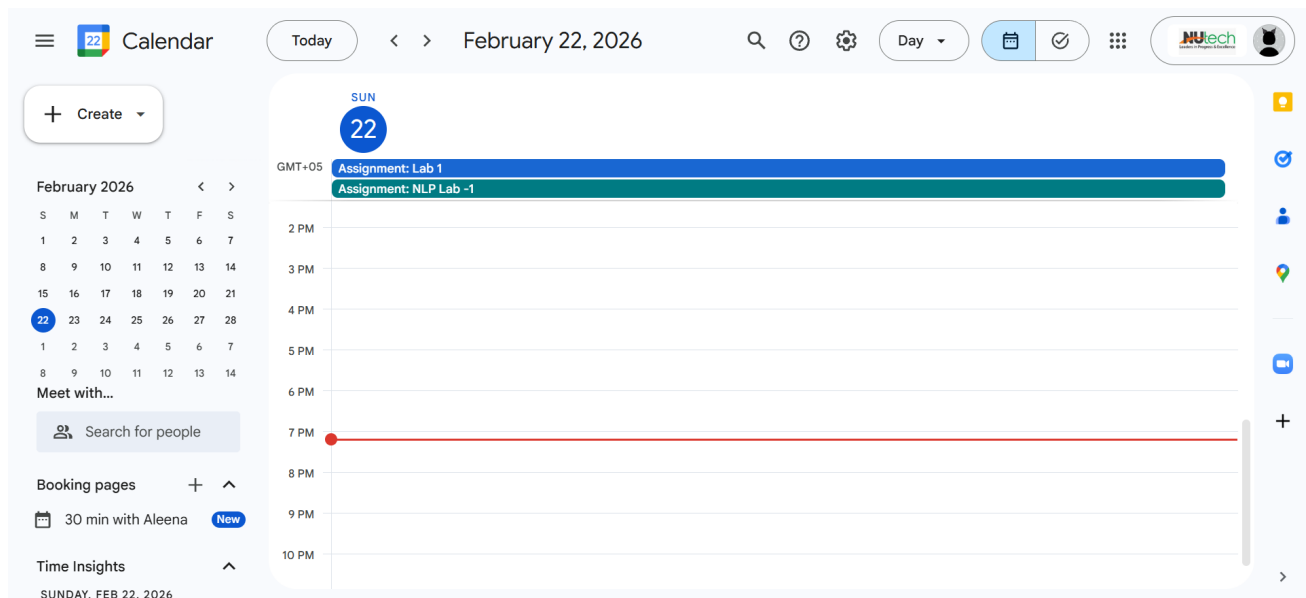


Figure 1: Google Calendar weekly view – a well-designed scheduling interface.

3.2 HCI Analysis – Strengths

Visibility: Google Calendar excels at visibility by displaying the current date, month, and all scheduled events in a single comprehensive view. The left sidebar shows a mini-calendar for quick date navigation and a list of calendar categories that users can toggle on or off. Events are color-coded by category (work, personal, holidays), making it immediately clear which events belong to which calendar. A moving red line across the day view always indicates the current time.

Feedback: When a user clicks on a time slot, Google Calendar instantly displays a quick-add popup with the time pre-filled. Drag-and-drop rescheduling provides real-time visual feedback as the event block moves smoothly with the cursor. After saving an event, a brief toast notification appears at the bottom confirming the action, complete with an “Undo” button for immediate error recovery.

Affordances: The prominent “+” button for creating new events clearly suggests its clickable purpose. Time slots on the calendar grid appear as interactive areas—hovering reveals a preview of the time being selected. Event cards are styled like physical sticky notes with rounded corners and subtle shadows, naturally inviting users to click and interact.

Consistency: Google Calendar maintains consistent design patterns across all views (day, week, month, year, schedule). Navigation arrows, color schemes, typography, and button placements remain uniform. The interface follows Google’s Material Design guidelines, ensuring familiarity for users of Gmail, Drive, and other Google products.

4 Case 2: Poorly-Designed Interface – Yale School of Art Website

4.1 Interface Overview

The Yale School of Art website (<https://www.art.yale.edu>) is the official website for one of the most prestigious art schools in the world. Despite belonging to Yale University, the site functions as an open wiki where any community member—students, faculty, staff, and alumni—can edit content and change the visual appearance. The result is a visually chaotic, inconsistent, and confusing interface that fails to communicate information effectively. The homepage features a distracting full-page background photograph, bright yellow highlighted headings, mismatched color blocks, and a cluttered layout that undermines readability and navigation.

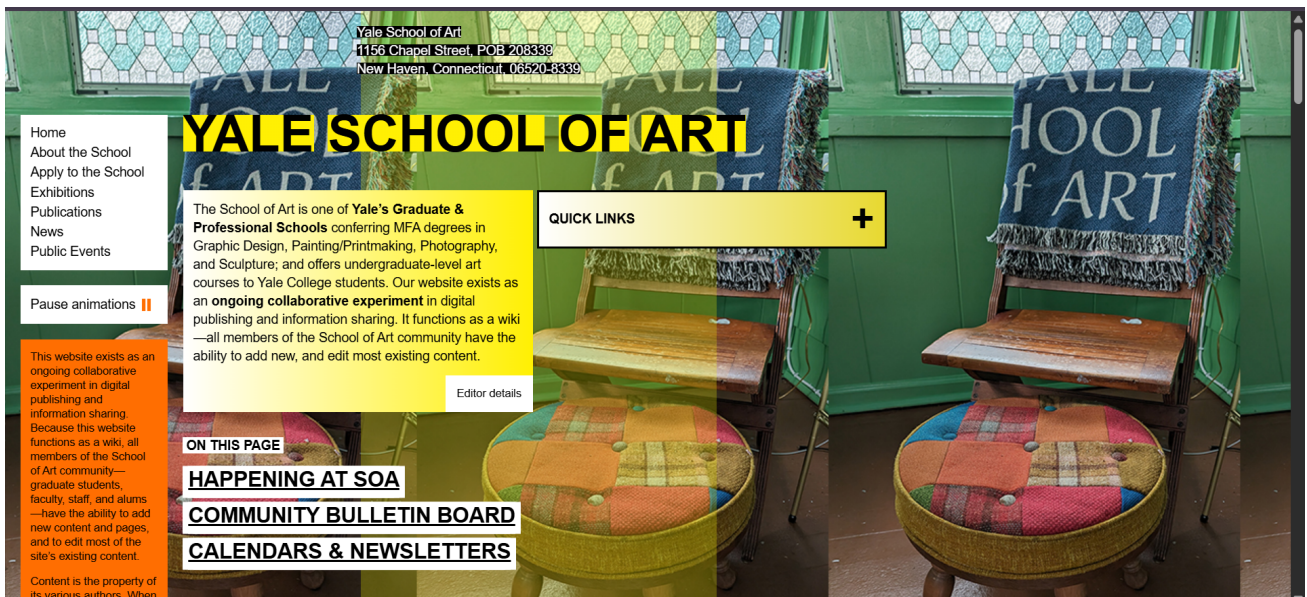


Figure 2: Yale School of Art homepage ([art.yale.edu](https://www.art.yale.edu)) – a poorly-designed interface with clashing colors, chaotic background, and no visual hierarchy.

4.2 HCI Analysis – Weaknesses

Poor Visibility: The homepage is dominated by a large, busy background photograph (showing colorful chairs and a storefront) that competes heavily with the foreground content. The title “YALE SCHOOL OF ART” is rendered in large black font directly over this photograph, making it partially unreadable. Important navigational elements and content sections like “Happening at SoA,” “Community Bulletin Board,” and “Calendars & Newsletters” are displayed as bright yellow highlighted blocks that clash with the background. Users cannot easily identify what content is available or where to find critical information such as admissions, programs, or faculty listings.

Lack of Consistency: Since the website is a wiki editable by the entire community, there is no enforced design system. Different pages use entirely different background images, color schemes, fonts, and layouts. The homepage alone mixes an orange sidebar description block, yellow highlighted navigation links, a white content area with standard text, and a gold “Quick Links” box—all with no cohesive visual relationship. Moving from page to page feels like visiting entirely different websites.

Poor Mapping: The left sidebar contains navigation links (Home, About the School, Apply to the School, Exhibitions, Publications, News, Public Events), but their placement overlaps with the background image and competes with the yellow highlighted “ON THIS PAGE” jump links in the main content area. Users face two competing navigation systems with no clear indication of which

one leads where. The “Editor details” and “Pause animations” controls are mixed in with navigational elements, confusing interface controls with content navigation.

No Affordances: Links on the page do not follow standard web conventions. The yellow highlighted headings (“HAPPENING AT SOA,” “COMMUNITY BULLETIN BOARD”) look more like highlighted text in a document than clickable links. The “Quick Links” box with a “+” icon provides no visual cue about what it expands into. Standard affordance cues like underlines on hover, button-like styling, or color changes on interaction are absent or inconsistent.

Poor Feedback: The website provides minimal feedback for user interactions. Hovering over links does not consistently produce visual changes. The wiki-editing functionality lacks clear confirmation messages. The “Pause animations” button controls the background image but provides no clear indication of its current state. There is no search functionality on the homepage and no breadcrumb navigation to help users understand their location within the site.

5 Figma Redesign of the Yale School of Art Website

5.1 Redesign Approach

Using Figma, the Yale School of Art website homepage was completely redesigned to address the identified usability issues. The redesign transforms the chaotic wiki-style layout into a clean, professional academic website with clear navigation, proper content hierarchy, consistent styling, and meaningful visual cues. The goal was to preserve the creative identity of an art school while making the interface functional, accessible, and user-friendly.

Figma Mockup Link: <https://www.figma.com/design/y3nzNUdP8ufiuChwsdfuXV/Untitled?node-id=7-262&t=enxPOUDY2HGn4LGJ-1>

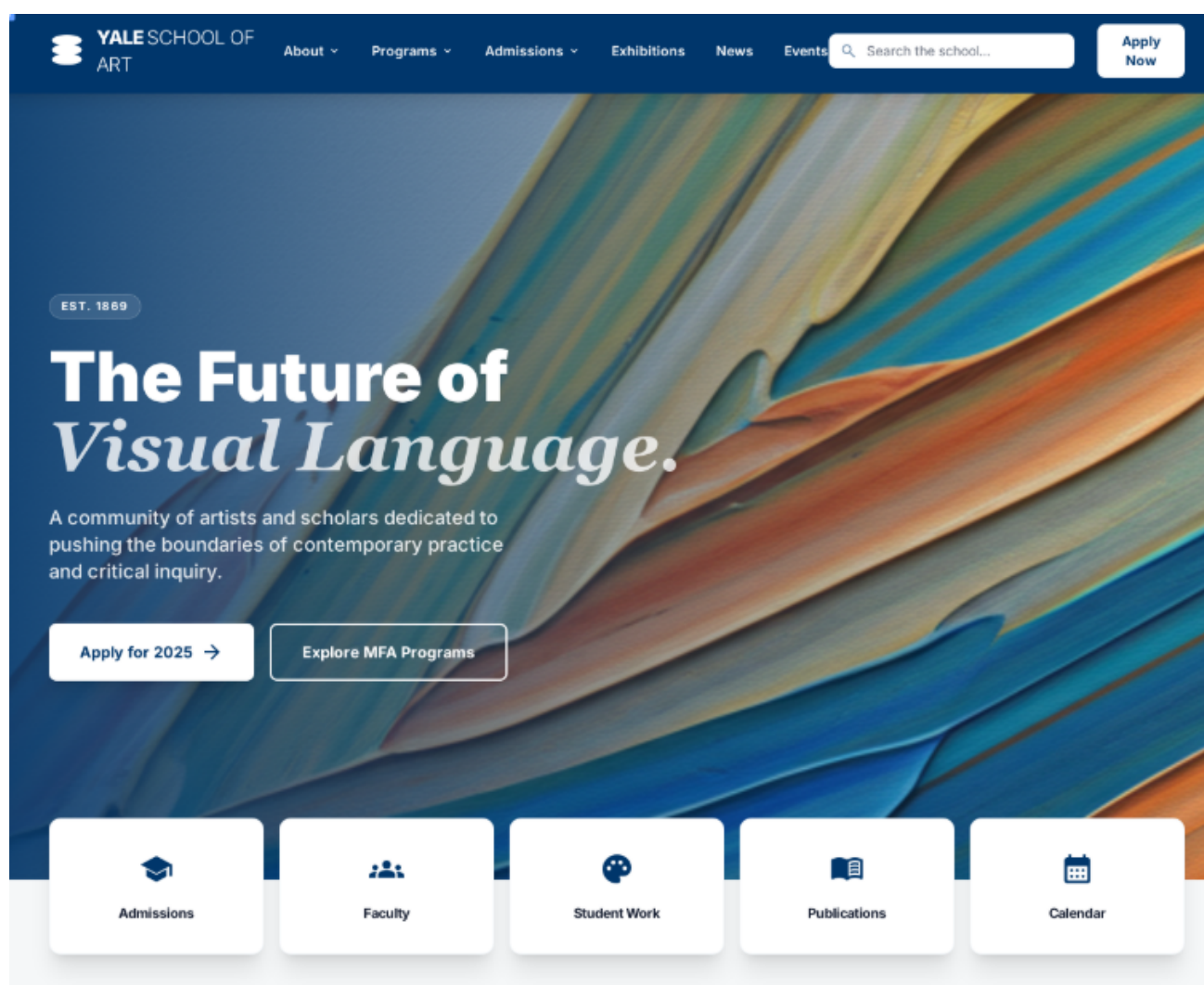


Figure 3: Figma redesign mockup of the Yale School of Art website.

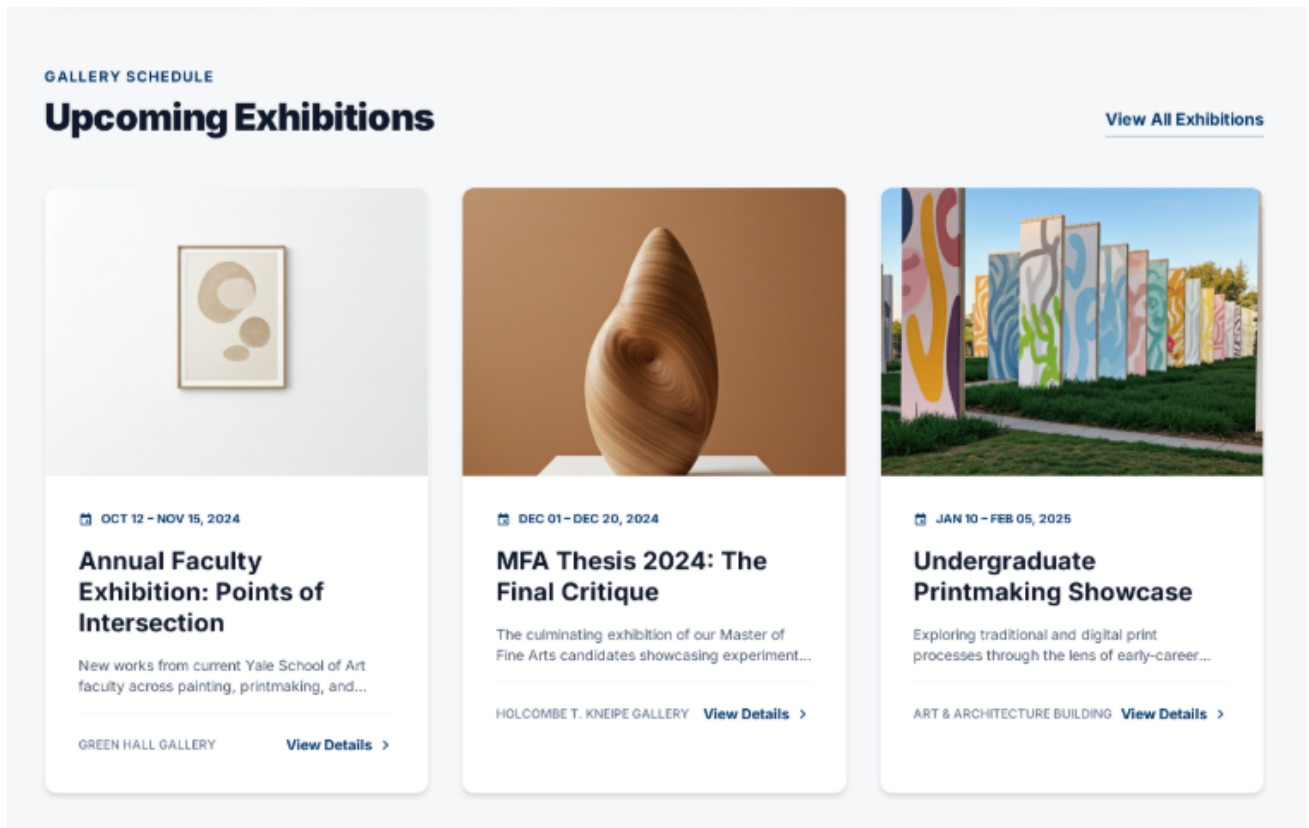


Figure 4: Figma redesign mockup of the Yale School of Art website.

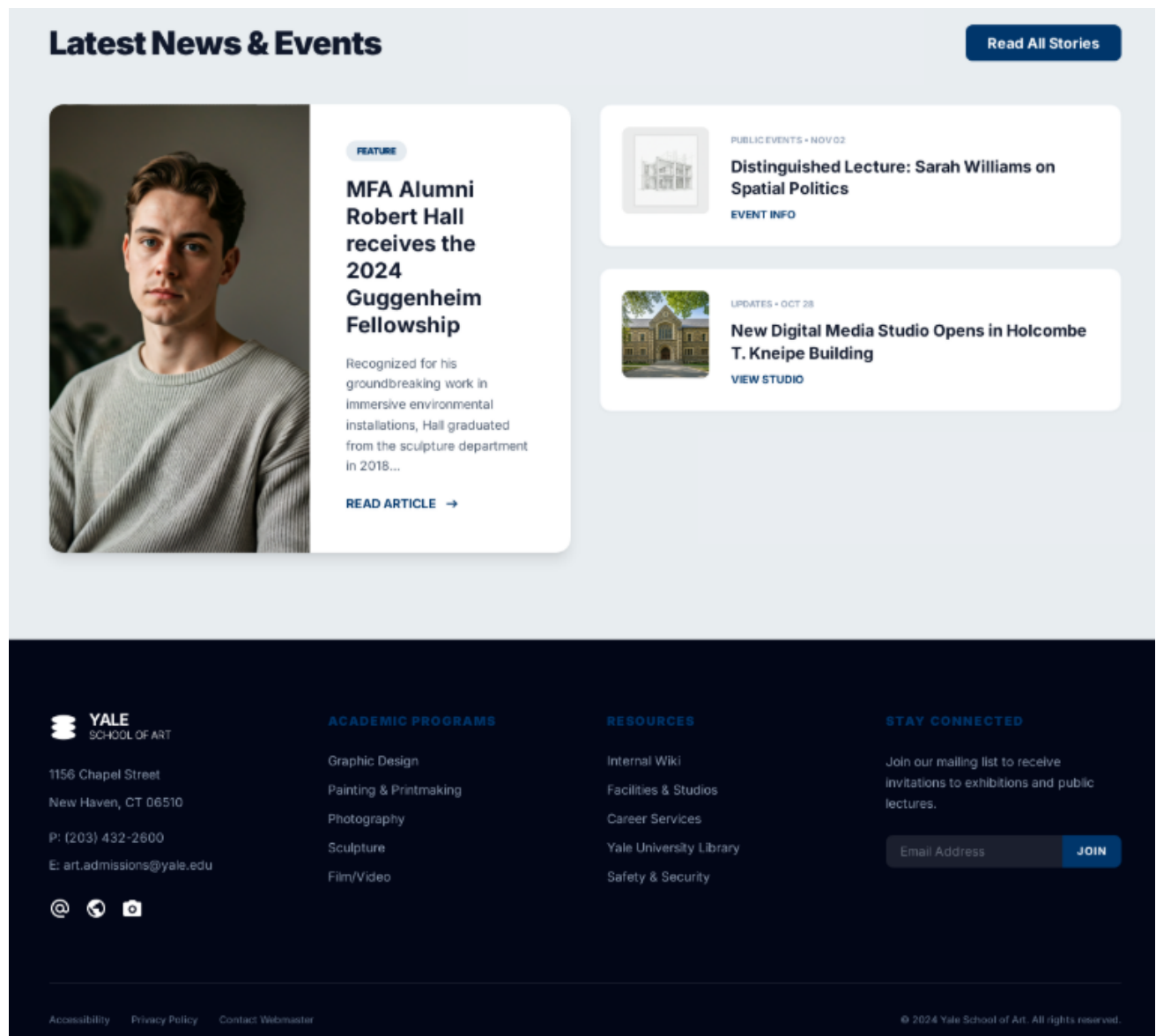


Figure 5: Figma redesign mockup of the Yale School of Art website.

5.2 Improvements and Applied HCI Principles

Table 2: Comparison of Original Issues vs. Redesign Improvements

nutDark		
Visibility	Busy background photo competes with content; yellow highlights create visual chaos; no content hierarchy	Clean white background with curated hero image; clear section headings; visual hierarchy using font size, weight, and spacing
Consistency	Wiki-editable pages have different backgrounds, fonts, colors on every page; no design system	Unified design system: Yale Blue (#00356B), white, light gray palette; single font family; standardized page templates
Mapping	Two competing nav systems (sidebar + yellow jump links); editor controls mixed with navigation	Single top navigation bar with clear dropdown menus; admin/editor tools separated into distinct toolbar
Affordances	Yellow highlighted text doesn't look clickable; Quick Links box unclear; no hover states	Styled buttons with hover effects; card-based sections with shadows inviting clicks; clear CTAs ("Apply Now," "View Exhibitions")
Feedback	No hover effects; no search; no breadcrumbs; no confirmation on wiki edits	Hover states on all elements; search bar in header; breadcrumb navigation; success/error notifications for edits
Error Prev.	Open wiki allows anyone to break design; no content validation	Role-based editing with preview before publish; content templates prevent layout breakage; revision history with rollback

5.3 Key Design Decisions in Figma

- **Top Navigation Bar:** Replaced the chaotic sidebar + yellow jump links with a clean horizontal navigation bar featuring the Yale logo, search bar, and clearly labeled dropdown menus (About, Programs, Admissions, Exhibitions, News, Public Events). This eliminates the dual-navigation confusion.
- **Hero Section:** Replaced the distracting full-page background photo with a controlled hero banner showcasing a curated artwork with a dark overlay for text readability. Includes a clear tagline and "Apply Now" / "Explore Programs" CTA buttons.
- **Card-Based Content Grid:** Replaced the free-form wiki content with structured card sections for "Upcoming Exhibitions," "Latest News," "Community Events," and "Quick Links." Each card has a thumbnail, title, date, and "Read More" link—all uniformly styled.
- **Consistent Color Palette:** Applied Yale's official blue (#00356B) as the primary color with white backgrounds, light gray (#F5F5F5) section dividers, and black text. Eliminated the random orange, yellow, and gold color blocks.
- **Typography System:** Used a single font family with a clear hierarchy: 36px bold for page titles, 24px for section headings, 16px for body text. Replaced the original's inconsistent font sizes and highlighted text.
- **Professional Footer:** Added a footer with contact info, address, social media links, and quick links organized in columns—replacing the scattered information placement of the original.

6 Conclusion

This usability detective exercise demonstrated the critical impact of HCI principles on user experience. Google Calendar exemplifies how thoughtful application of visibility, feedback, affordances, and consistency creates an intuitive interface used by millions daily without friction.

In contrast, the Yale School of Art website (art.yale.edu) illustrates how even a prestigious institution can suffer from poor usability when design control is decentralized. The open wiki format, while philosophically interesting as a collaborative experiment, results in visual chaos: clashing colors, competing navigation systems, absent affordance cues, and a complete lack of consistency that makes the site confusing and frustrating for prospective students, visitors, and community members.

Through the Figma redesign, we transformed the Yale School of Art's chaotic homepage into a clean, professional academic website that maintains creative energy while being navigable and accessible. This exercise reinforced that good design is not merely aesthetic—it is fundamentally about understanding human cognition and behavior to create interfaces that work with users, not against them.

7 References

- [1] Norman, D. (2013). *The Design of Everyday Things: Revised and Expanded Edition*. Basic Books.
- [2] Shneiderman, B., Plaisant, C., et al. (2016). *Designing the User Interface* (6th ed.). Pearson.
- [3] Nielsen, J. (1994). 10 Usability Heuristics for User Interface Design. Nielsen Norman Group.
- [4] Google Calendar – <https://calendar.google.com>
- [5] Yale School of Art – <https://www.art.yale.edu>
- [6] Figma Design Tool – <https://www.figma.com>
- [7] Figma Redesign Mockup – <https://www.figma.com/design/y3nzNUdP8ufiuChwsdfuXV/Untitled?node-id=7-262&t=enxPOUDY2HGn4LGJ-1>
- [8] Figma Tutorial – <https://www.youtube.com/watch?v=uQsyobT2Rv8&t=50s>